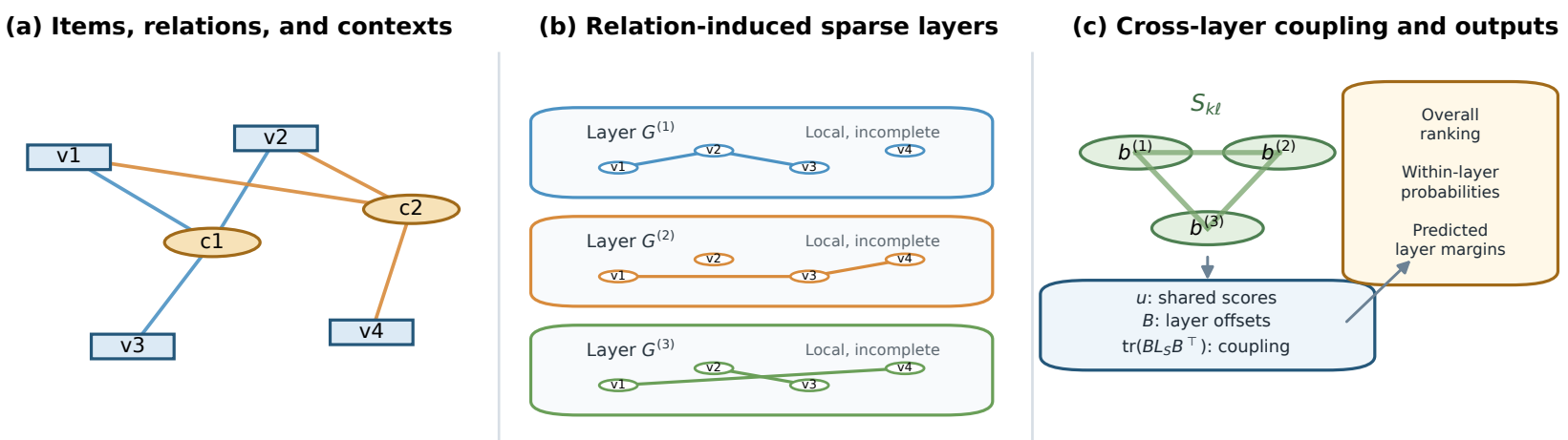
(a) Items, relations, and contexts
v1
v2
c1
c2
v3
v4
(b) Relation-induced sparse layers
Layer $G^{(1)}$
Local, incomplete
Layer $G^{(2)}$
Local, incomplete
Layer $G^{(3)}$
Local, incomplete
(c) Cross-layer coupling and outputs
$S_{k\ell}$
$b^{(1)}$
$b^{(2)}$
$b^{(3)}$
u: shared scores
B: layer offsets
$\mathrm{tr}(BL_SB^\top)$: coupling
Overall ranking
Within-layer probabilities
Predicted layer margins